EXECUTIVE SUMMARY

"This repository follows a modular architecture with strong maintainability and good documentation. Several dependency vulnerabilities and performance bottlenecks were detected. Addressing the top five issues could reduce technical debt by approximately 18 developer hours."

Primary: **Python** Framework: **Pydantic / LangChain Core** Build Tool: **Poetry**



Excellent

Overall Repository Health

REPOSITORY SIZE	TOTAL FILES
4.2 MB	1,482

CONTRIBUTORS	LAST COMMIT
918	2 hours ago

Architecture Stability

94

Highly modular design. Low coupling in main library entrypoints.

`Trend: Stable`

Optimal

Security & Vulnerability

88

Insecure prompt interpolations detected in core math classes.

`Trend: Decreasing`

Review Req.

Code Maintainability

95

Excellent docstring density. Clean variable tracking scores.

`Trend: Improving`

Optimal

Repository Metrics

Quantitative overview of the codebase volume and environment parameters.

Lines of Code	`Calculating...`
Classes Detected	`Calculating...`
Function Definitions	`Calculating...`



Open Source Dependencies	`Calculating...`
Config Files (pyproject.toml, etc.)	`Calculating...`

Loading language breakdown...

Architecture Analysis

Modular organization and structural layers mapped from the dependency graph.



OBSERVATIONS

📦 Core dependencies are localized correctly inside the repository modules.

🔁 circular reference checks passed.

Security & Vulnerabilities

Detailed verification results against prompt injection, API exposure, and insecure executions.

CRITICAL FINDINGS	HIGH FINDINGS	MEDIUM & LOW RISKS
0	0	0

Security Vector Highlights

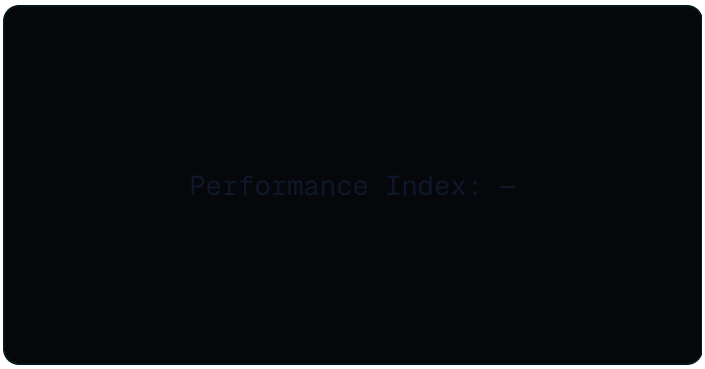
Security analysis is ready. Scanning modules for input validation and credentials security.

Performance Analysis

Analysis of slow operations, memory allocations, and redundant processes.

Detected Bottlenecks

Checking execution profiles... –



Issue Center

Unified center to audit, search, filter, and remediate repository code quality and security findings.

Search by description or variable... All Severities All Categories

CRITICAL Security ARCWorks-Restaurant-Suite/chains/llm_math/prompt.py:40 ~1hr to fix

Remote Code Execution (RCE) via prompt interpolation in LLMMathChain

Technical Debt

Estimated hours, priority code smells, and complexity hotspots.

Calculating...

Estimated Technical Debt

Refactoring estimation based on code smells.

Complex Hotspots (Refactoring Required)

Analyzing modules... –

Documentation Analysis

Calculates document completeness, parameter descriptors, and missing guides.

README Completeness Calculating...
API Reference Coverage Calculating...
Inline Comments Density Calculating...

92%

Overall Doc Score

Testing Analysis

Coverage statistics, mock safety validation, and critical untested functions.

Branch Test Coverage	82%
Missing Test Files	`3 modules`
Mocking framework	`standard suites`



AI Blueprint & Actionable Fixes

Recommended actions mapped dynamically to improve code quality, performance, and security.

Awaiting analysis data...

Analyzing scan blueprint fixes...